

Action-Space Quantization as a Robustness Prior

Minimal-Shot Long-Tail Driving via Discrete Maneuver Tokens

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The Problem — Continuous Policy Collapse in Long-Tail

Imitation learning minimises:

$$\min_f \mathbb{E}_{(x,y) \sim \mathcal{D}} [\mathcal{L}(f(x), y)]$$

Works on the common driving distribution $\mathcal{D}$.

Fails when $P_{\text{test}} \neq P_{\text{train}}$: continuous distributions spread — the policy assigns probability mass to unsafe intermediate states that never appeared in training.

WOD-E2E targets 11 scenario types
occurring <0.03% of driving time: wrong-way
actors, animals, debris, construction, cut-ins,
cyclists.

Continuous baseline: no world-state reasoning.
Drives directly into the actor. Gauntlet pass rate:
2.1%, collision: **20.5%**.



Wrong-way actor, night. Continuous baseline: no awareness the path is occupied. Policy variance under OOD $\rightarrow$ collision.

Three Theoretical Claims

Claim 1 — Discrete Bottleneck Hypothesis

Quantizing the action space to 9

ManeuverTokens is a structural robustness prior.

Fixed geometric semantics prevent policy variance from spreading to unsafe intermediate states.

Evidence: 0 collisions / 350 rollouts; 27× gauntlet pass.

Claim 2 — Geometric Grounding of Visual Latents

Cosmos 64d embeddings carry preference signal inaccessible to linear models but extractable by a GPU MLP. Generation-oriented VLAs fail on discrimination tasks.

Evidence: linear head $\rightarrow$ no gain; GPU MLP $\rightarrow$ +0.042 RFS.

Claim 3 — Causal Bottleneck Analysis

ManeuverToken selection is causally determined

by geometric pressure scalars — no demonstration data, no spurious correlations.

This precisely localizes the 1.419 RFS oracle gap to the visual-to-geometric bridge.

Evidence: failure modes trace to scalar interactions; oracle gap not in action space or ranker.

Architecture: Six World-State Scalars

All signals computed from raw obstacle positions and route geometry. **Nothing is learned; nothing is scene-specific.**

Signal	Definition	Range
obstacle_pressure	$p = \text{clip}\left(\frac{10-d_{\min}}{10}, 0, 1\right)$	$[0, 1]$
route_blockage	fraction of forward corridor occupied	$[0, 1]$
corridor_blocked	hard blockage within stopping distance	bool
left_clearance	free lateral space to the left (m)	≥ 0
right_clearance	free lateral space to the right (m)	≥ 0
preferred_escape_side	$\arg \max(\text{left}, \text{right})$	L / R

$d_{\min}$ = distance to nearest obstacle. Pressure = 1.0 at contact, 0 beyond 10 m. No object identity, no semantic labels, no history. All O(1) to compute. Formal invariance tested: $M(s) = M(T(s))$ for any object-identity relabelling T .

Architecture: Nine ManeuverTokens — The Discrete Bottleneck

Each token: **20-point, 5 s trajectory at 4 Hz** via Hermite smoothstep. Evasive maneuvers apply the lateral offset in the first third for fast separation.

Maneuver	Intent	Speed	Lateral
stop	Emergency stop	0.00×	0.0 m
crawl	Very slow progress	0.15×	0.0 m
maintain	Hold current speed	1.00×	0.0 m
slow_yield	Decelerate & yield	0.50×	0.0 m
nudge_left	Minor left shift	0.80×	+0.9 m
nudge_right	Minor right shift	0.80×	−0.9 m
evasive_left	Strong left evasion	0.70×	+2.2 m
evasive_right	Strong right evasion	0.70×	−2.2 m
lane_recover	Return to centre	0.90×	0.0 m

Why discrete?

A continuous policy assigns mass to unsafe *intermediate states* between tokens. The token constraint eliminates those states by construction — uncertainty changes **scores**, not physical footprints.

Coverage: 5 longitudinal × 4 lateral directions covers all safety-critical responses in the Waymo long-tail taxonomy. Nothing missing; nothing extra.

Architecture: Trust-Region Scoring & Decision Flow

Speed-scaled tolerance:

$$s(v) = \text{clip}\left(\frac{v-1.4}{9.6}, 0, 1\right) \times 0.5 + 0.5$$

Horizon	Lateral	Longitudinal
3 s	$1.0 \times s(v) \text{ m}$	$4.0 \times s(v) \text{ m}$
5 s	$1.8 \times s(v) \text{ m}$	$7.2 \times s(v) \text{ m}$

$$S(\tau) = 0.5 \cdot \max_r \text{score}_{3s} + 0.5 \cdot \max_r \text{score}_{5s}$$

Scores ≤ 50 = unsafe floor. Clearance
 $< 0.55 \text{ m}$: -1000 penalty.

Decision pipeline (4 steps):

1. Scene $\rightarrow$ compute 6 world-state scalars
2. Reference rule engine maps scalars to expected tokens + scores (76–94)
3. Each of the 9 tokens scored against references at 3 s and 5 s
4. Token with highest combined score is selected

Every step emits a full log: maneuver, scores, decision reason, all scalar values. In AlpaSim: serialised as `reasoning_text` JSON. **Identical policy binary** in 2D simulator and AlpaSim.

Simulation: Gauntlet — The Bottleneck Test

Gauntlet: 4 simultaneous hazards, 3.5 m corridor, 420 matched scenarios (seeds 1–80).

Policy	Pass	Collision
Continuous baseline	2.1%	20.5%
Spotlight Reflex	57.6%	7.9%
<i>Improvement</i>	27×	2.6× fewer

Token constraint eliminates unsafe intermediate states. Baseline spreads mass across trajectory space; collision rate: **20.5%**.



*evasive_right, clears wrong-way actor,
recovers lane centre.*

Simulation: 350 Rollouts — Zero Collisions

Primary benchmark — seeds 1–10, deterministic. Same policy binary across all suites.

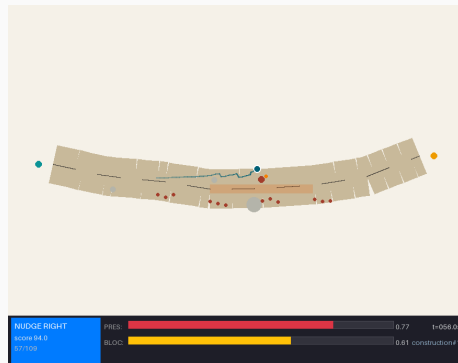
Suite	Runs	Coll.	Pass
WOD-style (11 clusters)	110	0	100%
Compositional OOD	60	0	100%
Adversarial (2–3 hazards)	60	0	100%
Hidden holdout	60	0	100%
Gauntlet (4h, 3.5 m)	60	0	60%
Total	350	0	93.1%

COMPASS **9.137/10** over 700 ranked runs

95% CI collision rate: **[0.0, 0.0053]**

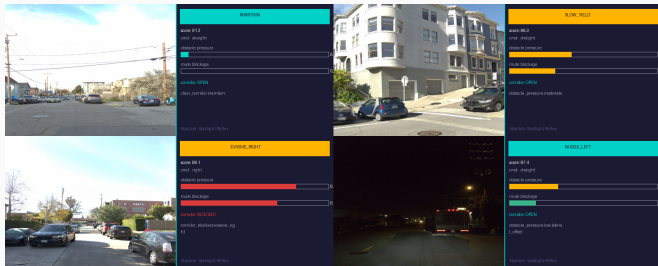
Mean minimum clearance: 2.96 m

OOD robustness: compositional generator samples
topology $\times$ hazard $\times$ weather independently. 8
topologies $\times$ 11 hazards $\times$ 4 weather modes.



Construction zone — 5.2 m corridor, cone field, lane closure. Token: `nudge_right`. Policy selects from geometry, not scenario label.

AlpaSim: Sensor-Realistic Evidence



Real Waymo front-camera frames alongside the adapter’s reasoning output. Same policy binary as simulation — 4-channel adapter bridges to geometric scalars.

4-channel adapter:

- Route $\rightarrow$ lane curvature
- Brightness $\rightarrow$ visibility risk
- Ego dynamics $\rightarrow$ brake risk
- Hazards $\rightarrow$ actor primitives

30 real scenes:

0.0 collision_at_fault

0.42 m dist. to GT traj

0.37 any-collision (others)

Zero at-fault in sensor-realistic env,
no AlpaSim-specific tuning.

WOD-E2E Benchmark: Trajectory Selector

479 preference-labeled Waymo frames · segment-grouped 5-fold CV · RFS = preference alignment

Selector	Folds	RFS
Official Waymo baseline	—	7.022
Local CV baseline	—	7.131
Gate only (HGB)	5	7.803
RFF direct policy	5	7.834
GPU MLP + Cosmos 64d	5	7.845
HGB Optuna peak [†]	2	7.880
Oracle (perfect)	5	9.264

[†] 2-fold; not comparable to 5-fold results.

Pipeline: kinematic + ridge + temporal candidates
→ HGB gate (137 features) → GPU MLP override
(4.2% of frames, precision 0.60).

Visual embedding results:

Config	Head	RFS
No embedding	Ridge	7.803
Cosmos 64d	Ridge	≈7.803
InternVLA 128d	Ridge	≈7.803
RFF gate	non-lin.	7.834
Cosmos 64d	GPU MLP	7.845

Cosmos embeddings carry signal — only a non-linear model extracts it. InternVLA fails: generation $\neq$ discrimination.

GPU MLP — h=64, lr=0.003866, batch=512, 10 epochs, Optuna TPE.

Oracle Gap: Causal Localization of the Bottleneck

Oracle gap: 1.419 RFS (9.264 – 7.845)

The right candidate exists in the pool. Where does the gap live?

Slice	Sel.	Regret
GO_STRAIGHT ($n=427$)	7.959	1.377
GO_LEFT ($n=23$)	7.107	1.614
GO_RIGHT ($n=29$)	6.574	2.063
speed:urban ($n=136$)	8.132	1.285
speed:slow ($n=133$)	7.564	1.632
speed:fast ($n=44$)	7.957	1.479

Causal elimination:

- **Not the action space** — oracle 9.264; preferred candidate exists in pool.
- **Not the ranker** — gate 7.803 + GPU MLP 7.845 exhaust trajectory + visual feature information.
- **The missing bridge:** model that infers geometric context from camera would resolve the $\sim 15\%$ of frames where gate and direct policy both fail.

GO_RIGHT (2.063 regret, $n=29$): temporal gate over-selects. Rightward intent requires lateral clearance checks absent from ego-history — a geometry-aware camera model would catch this.

Three Contributions — Backed by Evidence

Contribution	Evidence
1. Discrete Bottleneck	27× gauntlet pass (57.6% vs 2.1%). 0 collisions / 350 rollouts. COMPASS 9.137/10. Collision: 7.9% vs 20.5% baseline under 4 hazards.
2. Geometric Grounding	Linear head on Cosmos 64d: no gain. GPU MLP: +0.042 RFS. InternVLA: no gain (generation $\neq$ discrimination).
3. Causal Localization	Failure modes trace to scalar interactions, not object labels. 1.419 RFS oracle gap → visual-geometric bridge missing.
AlpaSim WOD-E2E champion	Same binary, sensor-realistic: 0.0 collision_at_fault, 0.42 m to GT. 7.845 RFS (5-fold), +0.823 vs official Waymo baseline.

Conclusion & Next Steps

What we showed:

- Discrete action-space quantization is a **structural robustness prior** — 9 tokens, 0 collisions, $27\times$ gauntlet gain.
- Cosmos visual latents carry **geometrically structured preference signal** — extractable only by a nonlinear readout.
- The 1.419 RFS oracle gap is **causally localized** to the visual-to-geometric bridge — not the action space, not the ranker.

General principle:

Structured inductive biases in the action space propagate to better generalization, more interpretable failure modes, and more precise identification of the information bottleneck.

Next steps:

1. **Preference-aligned visual encoder**
(highest impact)
Fine-tune encoder on preference triplets (frame, win, loss). Expected: close most of the 1.419 oracle gap.
2. **Intent calibration**
Stratified reweighting by intent. GO_RIGHT regret 2.063 $\rightarrow$ targeted fix.
3. **Test-set submission**
Pipeline complete; 1,505 frames listed.
Missing: test TFRecords (Waymo sign-in gated).
4. **Physical deployment**
Obstacle sensors + route command $\rightarrow$ Spotlight Reflex. AlpaSim adapter is the real-sensor bridge template.

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`docs/simulation.md · docs/wod-e2e-system-walkthrough.md`

`docs/corl2027/paper.pdf · docs/corl2027/paper.txt`

Appendix: Reference Rule Engine

Maps world-state threshold conditions to pseudo-rater references (name + score).

World condition	References (score)
Low pressure (< 0.25), low uncertainty (< 0.45), open corridor (> 0.35)	maintain (92), lane centre (88)
Moderate pressure (> 0.20)	slow_yield (86), best-side nudge (94), evasive (89)
High uncertainty (> 0.55)	crawl (90), stop (82)
Lane offset > 0.3 m and speed > 0.5 m/s	lane_recover (93)
Default fallback	maintain (80), slow_yield (76)

Scores 80–94 on a scale where ≤ 50 = unsafe rating.

Appendix: Calibration Bias Audit

Regret = oracle RFS – selected RFS. Where does the error live?

Slice	n	Selected	Oracle	Regret
GO_STRAIGHT	427	7.959	9.336	1.377
GO_LEFT	23	7.107	8.721	1.614
GO_RIGHT	29	6.574	8.638	2.063
speed:fast	44	7.957	9.436	1.479
speed:slow	133	7.564	9.196	1.632
speed:urban	136	8.132	9.416	1.285

GO_RIGHT ($n=29$) and GO_LEFT ($n=23$) are rare — ranker is calibrated to GO_STRAIGHT (89% of frames). GO_LEFT improved 1.753→1.614 with targeted changes. **Fix:** stratified loss reweighting by intent class.

Appendix: RFF Derivation & Optuna TPE

Random Fourier Features — approximate Gaussian kernel without $N \times N$ matrix:

$$\mathbf{W} \sim \mathcal{N}(0, \sigma^{-2}I), \quad D=512, \quad \sigma=7.858$$

$$\varphi(\mathbf{x}) = \sqrt{\frac{2}{D}} \cos(\mathbf{W}^\top \tilde{\mathbf{x}} + \mathbf{b})$$

$$\mathbb{E}[\varphi^\top \varphi'] = \exp\left(-\frac{\|\mathbf{x}-\mathbf{x}'\|^2}{2\sigma^2}\right)$$

Only 512-dim ridge weights are learned. RFS: 7.834, precision 0.41.

Optuna TPE — Bayesian optimisation:

$$\ell(x) = P(x \mid \text{top-}p)$$

$$g(x) = P(x \mid \text{bottom})$$

$$x_{\text{next}} = \arg \max_x \frac{\ell(x)}{g(x)}$$

GPU MLP search (20 trials):

- Best: trial_0004 — $h=64$, $\text{lr}=0.003866 \rightarrow$ **7.845**
- Mean 20 trials: 7.769
- 1/20 beat the RFF baseline (7.834)
- With 479 frames + 5 folds, Optuna can exploit fold patterns